Tides of Knowledge: ocean knowledge co-creation at the science-society-policy ecotone

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Knowledge is fluid, intertidal and moves across various contexts and realities. It spills over disciplinary and cultural walls, and moves with the rhythms of the world, in almost hydrological ways. Trying to contain it, without understanding this, is trying to contain a dynamic ocean without understanding its tidal customs. In this chapter we explore the possibilities for decolonizing ocean science through fluid and tidal movements toward ocean knowledge co-creation that is transformative, transdisciplinary and transgressive, and that expands research at the science-society-policy interface. In our experience coordinating and participating in a global transdisciplinary and transgressive ocean governance research network, we considered this approach essential to make progress towards SDG 14.a on increasing scientific knowledge and developing research capacity and marine technology, in order to improve ocean health and to enhance the contribution of marine biodiversity to the development of developing countries, in particular small island developing States and least developed countries. SDG14a is, in turn, crucial to enhance the conservation and sustainable use of the ocean and its resources by implementing international law (SDG 14.c) and ensuring access to resources for small-scale fishers (SDG 14.b) taking into account the inter-twined equity issues in ocean science and policy at different levels (Morgera, 2022). SDG 14a is also crucial to contribute to the realization of other SDGs. This chapter discuss how transformative, transdisciplinary and transgressive ocean research can help transform governance in the light of multiple SDGs, and reflects on the challenges, achievements and lessons learnt in this connection in the context of the One Ocean Hub– an international programme of research for sustainable development, working to promote fair and inclusive decision-making for a healthy ocean whereby people and planet flourish.

The chapter illuminates the importance of acknowledging the plural nature of knowledge, as well as inequities and evidence hierarchies in marine education, science, history, policy, and law, through a Global South lens. Using the metaphor of tides, this chapter then explores the lessons learnt in establishing the One Ocean Hub by bringing together coastal people, established and emerging researchers, decision-makers, civil society, and international organisations to value, and learn from, different knowledge systems and voices with a view to decolonising ocean science. We undertake this by drawing from transgressive learning (T-learning) theory (Lotz-Sisitka et al, 2015 and 2016; Lotz-Sisitka, 2024), which posits that through boundary crossing, iterative and transgressive social learning practice builds transgressive research praxis that spans not only different disciplines, but makes room for, and builds entrances and exists between, onto-epistemological renderings of our world. Lotz-Sisitka (2024:1) has recently referred to this approach as a "politics of transgression in environmental education research".

Decolonising ocean science, we argue, is a collaborative, iterative and transgressive co-construction process, which involves diverse perspectives converging around shared concerns. In

the examples we cover, T-learning processes incorporate partners from various organisations, including social movements, NGOs, academia, policy institutes, and community groups. This approach created an inclusive transgressive knowledge network (Lotz-Sisitka, 2024) that recognizes the importance of contextual diversity and multiple forms of knowledge, particularly regarding how sustainability issues impact communities.

This chapter will then reflect on five years of contextual examples of approaches and methods for transformative, transdisciplinary and transgressive collaborative research and action on SDG14 and other SDGs, as prototypes for transformative ocean governance. Key features of such a transformative science-society-policy ecotone include iterative co-design, co-identification of matters of concern and care-full co-creation and assessment protocols, the development of a shared Code of Practice, ethics of care that go beyond do-no-harm, the creation of entrances and exits between knowledge systems (de Sousa Santos, 2007), and sensitivity to different spatial and temporal scales. By this we mean, finding alternative pathways between disciplines and ways of knowing, being and doing. The chapter will conclude with seven key messages for regenerative, inclusive and collaborative meaning- and decision-making on the ocean, and its crucial role in nourishing multi-scalar partnerships (SDG 17) and protecting everyone's human rights (SDG 16) that depend on a healthy ocean (Bennet et al, 2024).

An intertidal ecotone

The intertidal zone is a unique, biodiverse, ecologically dynamic and fluctuating ecosystem that is shaped by the meeting of land and sea, the meeting of terrestrial and marine realms (SDGs 14-15). This dynamic land-sea interface is characterised by constant changes due to tidal fluctuations, which bring new nutrients and transfer organisms across the zone. This meeting of ecosystems and/or unique biomes, is referred to as an ecotone, which in the field of ecology refers to the interactions between neighbouring biological communities in a transitional area (Eppelsheimer et al, 2015). This liminal area shares characteristics of both marine and terrestrial communities and contains unique and rich biodiversity that cannot be found in either realm/ community alone due to the dynamic interaction between them (ibid, 2015). The concept is generative and compelling, and has been adopted by environmental humanities, which uses 'ecotones' as a generative theme for understanding dynamics of when territories, languages, knowledges and heritages touch, overlap and spill into each other (Denning, 2021) The crossing of boundaries and establishing of new, unique yet common ground, that is fluid and tidal, and is based on diversity and difference creates interesting synergies similar to those identified by biologists and geographers who work in and with ecotones. Ecotone is used in this chapter as a figurative[1] map for understanding the collaboration, interaction and influence between plural knowledges and ocean humanities and sciences (ibid, 2015), which is essential to decolonize ocean science.

Organisms making homes in the intertidal ecotone must be able to survive changes in moisture, temperature, and salinity and withstand strong waves. Similarly, coastal communities, economies,

[1] We use 'figurative' here in relation to the post-human philosophies who see "figuration" as a means to escape 'representation', referring to how humans and non-humans are portrayed within social or cultural contexts, challenging human-centric views. It involves deconstructing anthropocentrism, exploring alternative representations, and acknowledging the agency of non-human entities, fostering a reimagining of human-nonhuman relationships and identity.

and political ecologies living in times of the triple environmental crises (climate change, biodiversity loss and toxic pollution) (Passarelli et al, 2021; UNFCC, 2022) and rampant capitalist growth that spills over and affects their daily lives; must survive and cope with strong political currents, unforeseen economic waves, and unpredictable relational/intersectional cultures. As such, we articulate a fourth threat of a social inequality crisis (Österblom et al, 2023; Hornidge et al, 2023; Kirkman et al. 2023), i.e. the uneven distribution of resources and opportunities among people, which often results in poverty, hunger, and social injustice. This crisis amplifies the effects of the other two crises, as those who are most vulnerable to the impacts of environmental degradation are often the most marginalised and disadvantaged communities. Social inequality is also increasingly seen as the root cause of the triple environmental crisis (Puentes Riano, 2024).

We use this metaphor of the intertidal zone carefully, aware of the dangers (and harms) inflicted on society by trying to compare ecological dynamics with human ones. And so, we use this metaphor, not as a comparison, but a speculative figuration (Haraway, 2016), to offer a narrative thread that supports our portrayal of ocean knowledge-co-creation for decision making, in our experience, and see the figurative imagining as a translation protocol, between disciplines. We see this as useful not just for knowledge co-creation, but also as an opportunity to see the need for greater fluidity in governance that currently tends to see the making of decisions on the ocean and on land separately.

Knowledge communities operate in sovereign ways, that must navigate and live within power and class gradients, where culture, class, history, politics and economics spill over and shift according to many variables. In the Global South particularly, communities have had to navigate shifting colonial whirlpools of genocide, epistemicide, cognitive exclusions, environmental racisim, amongst other violent injustices (Martin, 2024; Peers, 2024; Kirkman et al 2023; de Sousa Santos (2015).

Our intertidal ecotone can also be used as a generative theme to describe the convergence of various disciplines, ways of thinking and doing, and onto-epistemological entanglements within the marine science/society/policy interface. As in the north, this intertidal place-making assists in describing and understanding the ecologies of knowledges (de Sousa Santos, 2017) that exist in the ocean decision-making in Indigenous territories globally, and offer insights towards inclusive, just and restorative approaches to ocean governance. Through the One Ocean Hub, a community of practice among researchers in the Global South and the Global North, which functions as a collaborative research ecotone, examined and contemplated the blue economy, the legacies of colonialism, decision making processes based on exclusionary scientific principles, and the cosmologies of Indigenous peoples and small-scale fishers in ocean decision-making.[2]

The One Ocean Hub aimed at creating a collaborative transdisciplinary research incubator and innovation collaboration between Global North and South researchers that supports transformative and transgressive learning (learning that is boundary crossing of epistemologies,

[2] www.oneoceanhub.org. The One Ocean Hub is a collaborative research programme for sustainable development project funded by UK Research and Innovation (UKRI) through the Global Challenges Research Fund (GCRF) (Grant Ref: NE/S008950/1) for the period 2019-2024. GCRF is a key component in delivering the UK AID strategy to tackle the Sustainable Development Goals.

cultures and systems) (Lotz-Sisitka et al, 2015; 2016; Peters & Wals, 2016; Bengtsson, 2019; Wals, 2021) as a contribution to marine governance through the understanding of the interconnectedness of ocean ecosystems of knowledge. The One Ocean Hub has, like intertidal marine ecosystems, become a nursery for innovative approaches to ocean research, planning, decision-making and governance.

Furthermore, considering that all oceans are connected, our metaphorical navigation also draws from deep-sea ecology, with a healthy attentiveness to the deep unknown, hidden and unforeseen outcomes and possibilities that sit in the touching of knowledge territories, and how they connected or disparate, yet in some cases are somewhat out of reach. While we know deep-sea ecosystems play a critical role in the health of the ocean, humanity and the planet, much of the vast deep-sea realm is still uncharted and unexplored by science. So too, in our navigation of plural ecologies of knowledge in the ocean-related decision-making arena, there is much below the surface, significant blindspots shaped by legacies of colonialism, that still need to be addressed (Whittingham & McGarry, 2024).

An attentiveness to hidden deep ecologies of knowledge and the emotional, spiritual and intangible cultural heritages associated with them, we believe, is critical in decolonizing ocean science and understanding the science-society-policy interface in marine governance, not just in the global South, but internationally. Here we celebrate those migratory species that, like in deep-sea ecologies, vertically migrate between knowledge ecosystems, bringing with them nutrients and stirring up the waters. Their bioluminescent framings of knowledge co–production across cultures of class, race, discipline, economies, and politics offer new light in dark times.

Why do we need transdisciplinary, transformative and transgressive research in the ocean space?

We unpack the call for transformative change and transformative ocean science at the international level, and the relevance of transdisciplinary and transgressive research to decolonize ocean science and prototype transformative science-society-policy ecotones. The need for transformative change to address the current planetary crises has been widely recognised internationally, from the 2030 Agenda for Sustainable Development to the 2022 UN Ocean Conference and the Kunming-Montreal Global Biodiversity Framework 2021-2030. Transformative change encourages the shifting away "from the technocratic and regulatory fix of environmental problems to more fundamental and transformative changes in social-political processes and economic relations" towards integrative, inclusive, adaptive, and pluralist approaches that better address both the indirect and direct drivers undermining sustainability (Visseren-Hamakers & Kok, 2022). Such changes are occurring in higher education and in efforts to decolonise marine and environmental sciences, led by powerful African feminist scholars like Aaniyah Martin (2024), Joanne Peers (2024), Black Feminist scholars like Alexis Pauline Gumbs (2021), and the field of critical African Feminism and Indigenous Feminisms (Tamele & Knowles, 2021) producing anti-racist, pluriverse and transgressive forms of science and research (Behl et al, 2021; Whittingham & McGarry, 2024). African marine sociologists are shifting ocean governance and policy to include customary Indigenous systems, like Philile Mbatha (2022), along with transformation in marine science itself (Moore, 2022). This is seen as essential to include underrepresented types of knowledge and

empower those represent transformative sustainability values ensure, but whose interests are currently not being met, in the development of solutions that can have sustainable impacts across scales and sectors (Visseren-Hamakers & Kok, 2022). Such transformation entails in ocean (but also legal and other) sciences a flexible and iterative design and multiple approaches enabling learning, experimentation, reflexivity, monitoring and feedback (Erinosho et al, 2022).

In response, the UN Decade of Ocean Science for Sustainable Development calls for "transformative science", as inclusive science that spans the marine and social sciences, is co-designed and co-produced in partnership with a variety of stakeholders and knowledge holders (including Indigenous peoples and local knowledge holders), uses Agenda 2030 as framework for priorities and emphasises solutions for governance (Lombard et al, 2023). This definition, however, shies away from the challenges that *transformative* science needs to face as part of the "politics of transformative change" (Visseren-Hamakers & Kok, 2022). This also requires preventing a shifting of the burden of response onto the vulnerable, paying attention to social differentiation, through the lens of non-discrimination, and addressing issues of power and legitimacy (Visseren-Hamakers & Kok, 2022). This is particularly challenging as justice (SDG 16) and human rights in relation to the ocean are less visible than on land, and research in this area is still incipient (Erinosho et al 2022; Morgera, 2024c; Bennet et al, 2024).

Within the Hub, we explored a wide variety of transformative research approaches that are used at different scales and degrees. All were in service of building 'entrances and exits' between knowledge systems, allowing for areas of work to be sovereign but also in solidarity with others working in the ocean governance space and in support of ocean communities. In this context, transformative research refers to research that shifts or breaks existing scientific paradigms, encourages transdisciplinary collaboration and is used to understand, bridges and transcends beyond disconnects across the science/policy interface. In turn, *transdisciplinary* research refers to knowledge co-production not only across different academic disciplines, but also across different knowledge systems and is seen as a precondition for transdisciplinary governance, i.e. the inclusion of underrepresented types of knowledge, for supporting integration, inclusion and learning in ocean affairs for transformative change (Visseren-Hamakers & Kok, 2022). Furthermore, transdisciplinary research aims to co-design and create knowledge systems and protocols with Indigenous peoples, small-scale fishers, rural women and other local knowledge holders, human rights and customary rights holders in research practices to catalyse processes of change and contribute to solving societal problems (Morgera, 2024c; Erinosho et al, 2022). Its explicit aspiration is to engage and bring about transformation in ocean science-society-policy interface.

While not all researchers in the One Ocean Hub are transdisciplinary researchers, just as within the ecotone some species are exclusively terrestrial while others are exclusively marine, their impacts, contributions, wastes, etc do, however, influence the ecotone and its diversity and nutrient cycling. So too, in a transdisciplinary research hub, some researchers were not at first glance conducting transdisciplinary research, yet their contributions did influence and support various aspects of it. For example, some deep-sea scientists realised in time, that their research could influence transgressive learning and capacity development in the Global South and being connected with Indigenous beliefs in the role of the ocean in the global water cycle (McGarry,

2023), having an impact on education (SDG 4) and much-needed transformations in deep-sea science (Sink et al. 2021).

Transgressive learning, in turn, is a radical, optimally disruptive (Wals, 2021) yet convivial form of social learning that goes beyond knowledge reproduction and challenges normative foundations through crossing knowledge boundaries, as well as other societal boundaries (Lotz-Sisitka et al, 2016). While transformative learning focuses on critical cognitive reflection and self-reflection as means of transforming perspectives and practice, transgressive learning (or T-learning) is characterised by a break with continuity, where held perspectives and assumptions are momentarily left behind to explore radically different ways of being, knowing and doing (Bengtsson, 2019). Transgressive social learning (Lotz-Sisitka, et al, 2016) uses practical co-designed pedagogies and techniques to disrupt and disengage from existing systems of practices and normopathies, in order to create new spaces for transformative research and learning. In some situations, T-learning can do this through arts-based practice and Indigenous-led meaning-making, that offer 'suitable strange' (McGarry, 2022) learning possibilities that can stimulate transformation in knowledge making, and shift normative practices and systems. It is a critical approach to learning that seeks to create responsive, accessible and meaningful knowledge and practices that are not bound by existing frameworks or paradigms. T-learning within the One Ocean Hub was associated with social and environmental justices, and was used to facilitate transformative research design and collaborative analysis and action across social justice, sustainability, and cultural diversity in relation to ocean science and governance. For example, a theatre production on different ocean knowledge systems (marine sciences, Indigenous knowledge, small-scale fishers' knowledge) and multi-generational experiences of removals from the coast and exclusion from ocean-related decision-making processes along the West Coast of South Africa revealed unaddressed historical and ongoing injustices in marine conservation and management (Erwin et al, 2022). Equally, this theatre production surfaced possibilities for bridging the academic-activist divide, including scholar-activists who had established relationships with small-scale fishing communities and social movements, which led to solidarity building across academic disciplines in service of small-scale fisher movements along the whole coast of South Africa (Pereira and Erwin, 2023).

Both transformative research and transgressive social learning involve aspects of questioning and challenging existing structures and systems, and within the One Ocean Hub, this collaborative research approach encouraged individuals and communities to explore new possibilities and ways of being and doing. As this process requires rethinking and challenging norms, it can be demanding, as it requires individuals to step outside of their comfort zones and embrace uncertainty and complexity. This has resulted in One Ocean Hub researchers using transgressive approaches to greater and lesser degrees, according to the freedoms and obstacles in different contexts to cross boundaries or transform approaches, helping individuals to develop self -critical and transformative skills, practice *response-able research*, solidarity, grow dynamic knowledge networks, and build political rigour (Temper et al, 2019). A 'response-able research' approach emphasises ethical responsibility, accountability, and responsiveness to the needs and concerns of various knowledge holders, particularly individual and groups in marginalised or vulnerable situations. This concept extends beyond traditional notions of research ethics and incorporates a deeper commitment to social justice and transformative action. In "response-able" research, researchers recognize their role in shaping knowledge production and actively strive to engage with

communities in a collaborative and respectful manner. This involves listening to diverse perspectives, acknowledging power dynamics, and prioritising the voices and experiences of those directly affected by the research topic. We have also witnessed how this approach can offer not only researchers, but also policy makers and judges new ways of engaging with new ideas and perspectives, and take action to create positive change.

The need to address power dynamics and politics of knowledge production in ocean science and ocean governance

To undertake transformative, transdisciplinary and transgressive ocean research, it is necessary for marine science to be(come) conscious of itself and its role in addressing (or ignoring and therefore entrenching) the inequalities and inequities that exist at the science-society-policy interface, particularly in the context of Global North/South imbalances, and how these may be obvious or insidiously hidden in plain sight. An example of the subversive can be seen in Max Liboiron (2021) work on "Pollution is Colonialism", which highlights the pervasive presence of pollution as a form of colonial occupation, which reflects and exacerbates existing colonial structures, reinforcing inequalities in environmental justice (in some cases environmental racism), as pollution continues to disproportionately affect Indigenous territories over generations.

The first step towards transformative, transdisciplinary and transgressive ocean research is engaging with the bodies of research (Science Technology, and Society scholarship, Environmental Justice scholarship) that have documented and analysed the power dynamics and politics of knowledge production in science. Common assumptions that scientific advances are necessarily beneficial or that scientific knowledge is objective need to be critically assessed. Science Technology, and Society research has unveiled that science is socially construed. Although in principle it is expected to be disinterested, based on peer review and on a culture of sharing among scientists, in practice it is often marked by competitiveness, secrecy and hidden interests and agendas, with the growth of interdisciplinarity destabilising the checks of established disciplinary standards and community of peers (Jasanoff, 2011).

The power dynamics at play in science have also been increasingly revealed. Impacts of neoliberalism on scientific research practices have been detected, such as the diminution of public funding, the narrowing of scientific agendas on the needs of commercial actors, and the intensification of intellectual property rights impeding the production and dissemination of scientific findings (40 *Social Studies of Science*; A/HRC/55/44). Indeed, even when there is an explicit agenda of collaboration, questions, framings and priority species or ecosystems to be studied are framed by Global North researchers, who are fund-holders, and their values and priorities for career progression and international recognition, which may be radically different from the thematic and spatial research priorities for enhanced ocean governance in the Global South. In addition, Environmental Justice scholarship underscores that the definition of science itself has led to exclusion and marginalisation: the self-connotation of Western European science (as captured in peer-reviewed academic publications) has contributed to the denigration of Indigenous knowledge and local knowledge (as captured in storytelling), although both are social constructs and in fact "modern" science can be historically seen as an increasingly standardised form of local knowledge [Vermeylen et al, 2018]. Concerted attention must therefore be paid to cognitive justice

- the equal treatment of all forms of knowledge (Leibowitz, 2017) recognising that the ways in which people come to know, use, and care for the living world, which are locally and culturally unique, offer opportunities for meaningful ecological care practices and should therefore be engaged as part of a dialogue between knowledges (Erwin et al, 2022).

This is linked to another dimension of justice – recognition, which speaks to the need to make visible the previously invisible (or privileged irresponsibility or ignorance: Bozaelk, 2023). In situations of environmental or cognitive injustices, certain communities and individuals have been unfairly affected by biological science and decision-making systems that have historically excluded the distinctive perspectives and experiences of those negatively affected. For instance, in the World Heritage Site of iSimangaliso in South Africa, Indigenous knowledge and land-freshwater-coastal ecosystem management practices were dismissed by Western fisheries science, leading to the displacement of Indigenous communities and the lack of recognition of their tangible and intangible cultural heritage in the international designation of the site (Empatheatre: Umkhosi Wenala play, 2023; Mbatha, 2022).

Recognition thus serves to call attention to diffuse realities of domination and oppression (patterns of non-recognition and disrespect of certain groups, stereotypical public and cultural representations of these groups, denial of their rights and denigration of their ways of life) that causes unjust distribution of resources and benefits, and unjust processes of participation in relevant decision-making practices (Young, 1990). One key area of ocean governance where recognition is crucial is the interaction between State laws and the customary laws of Indigenous peoples, small-scale fishers and local communities and, through that, recognition and protection of these communities' contributions to the conservation and sustainable use of the ocean (Christoffels-DuPlessis et al, 2022). Such recognition can only arise from decolonizing ocean science as a precursor to decolonizing ocean law and governance, by expanding science to include on an equal footing Indigenous science and plural approaches towards knowledge co-production, in rigorous ways (Temper et al, 2019), that can support decision-making processes that are enriched by, and do not discriminate against, different knowledge systems.

Transformations in marine science, therefore, requires a more radical approach than simply creating more spaces for Indigenous knowledge holders, women and people of colour in science (McKitrick, 2020). Rather, the vital task of decolonizing science requires to shift away from the current evidence hierarchies that prioritise western scientific framings over other local and Indigenous forms of knowledge and moving away from integrating Indigenous knowledge and local knowledge into established western scientific knowledges (which is being done currently in clumsy and sometimes offensive ways). Rather, it is about centering knowledge co-production in Indigenous knowledge practices, methodologies, customs and protocols. An illuminating example is that most Indigenous meaning-making practices are inherently relational ontological systems, whereas sustainability science is constantly trying to build novel systems thinking research. What helps us here is a transgressive and transdisciplinary approach that challenges these hierarchies and recognizes the importance of pluralistic knowledge co-production (Lotz-Sisitka et al, 2016). The first step is realising that research questions and hypotheses cannot be set by researchers alone, but must be co-defined with Indigenous peoples, and other customary rights and knowledge

holders, thereby starting to develop a generative partnership for transformative science and governance.

A decolonial marine science will require a regenerative and living push-and-pull dynamic between ways of knowing, being and doing. It will need to develop new high and low tidal markers, where changes can be made, through iterative shifts in power dynamics, dismantling of evidence and knowledge hierarchies, and collaborative meaning-making practices. Returning to our tidal ecotone metaphor, this tidal zone is a place of both decay and new regenerative life. Old realities need to wash ashore, be examined, dismantled by the intertidal ecological communities, and fed into new relational practices, systems and resources.

Work Package Zero

Allowing for this back-and-forth dynamic, and building on the iterative logics of the ecotonal realm of and its regenerative capacities, requires research co-design from the very outset. This is not always possible at the stage of applying for research grants: in the case of the One Ocean Hub, for instance, extremely tight deadlines were set to create an unprecedented programme of collaboration in terms of scale, partnerships and methodologies. For that reason, the grant application acknowledged how the time limitation fundamentally undermined a genuine possibility to co-develop the research programme across diverse knowledge holders, and included, as a response, an innovative concept – a Work Package Zero (WP0). This had funding and time implications, as we decided to set aside at least the whole first year of research to re-designed all initially research plans developed by academic researchers (albeit many with pre-existing, decade-long relationships with ocean knowledge holders) with, and truly in service of, Indigenous Peoples, small-scale fishers, rural women and other ocean-dependent communities.

A combination of arts, anthropology, sociology, history, political ecology, science and technology studies, and transgressive learning for sustainability were employed to move beyond previous interdisciplinary endeavours that did not engage sufficiently with agency, power, identity, space and time, ideological tensions and 'invisible' justice issues as much in ocean research endeavours, as in ocean governance. Relevant methodologies were combined in the iterative WP0 that would provide ongoing engagement with ocean-dependent communities, to co-identify matters of concern in relation to access to marine areas (SDG 14.c), gender equality (SDG 5) and cultural heritage (SDG 11). This initial phase also served to ensure ethics of representation in research outputs, and to empower those researchers who were not working directly with communities to experience their lived experiences and connect with their knowledge systems. All this was essential to allow the whole research programme to engage in transdisciplinarity. This, in turn, was necessary in moving away from more limited approaches to "stakeholder engagement" aimed at obtaining "social acceptability" for pre-conceived ideas and approaches developed by researchers, which is typical not only of ocean research but also broader sustainability governance.

Instead, arts and social sciences created and studied, self-reflexively, the process of getting to know, developing dialogue and collaboration, and learning from ocean-dependent communities (Erwin et al). This served as a phase for many researchers to see their position in the transdisciplinary ecotone, learn to have a genuine regard for other related knowledge systems, and identify any

unconscious bias, systemic discrimination of diverse knowledge holders, or preconceived idea of the research aims and approaches that could work against justice (SDG 16). This also raised the awareness of researchers about occurrences in which human rights had already been violated in ocean science and/or governance and start considering spaces in ocean research for the protection of human rights as envisioned by ocean-dependent communities.

WP0 required asking and exploring sometimes difficult questions with collaborative investigators, their partners and Indigenous peoples and other customary rights holders in each country, such as: what do researchers think they know and what don't the know (data sets, existing research, challenges)? What do researchers understand about whose voices are missing in ocean-related decision-making and who should be involved in ocean governance? Such critical self-examination was one of the key rationales for WP0, as often limited understanding of Indigenous knowledge and local knowledge by researchers can lead to inaccurate assumptions about knowledge gaps. Other questions that were explored were: what has gone wrong in the past (in research and historic interventions)? How do we avoid past errors and redress them? And what are the matters of concern (issues, struggles, and locally recognized knowledge gaps) across all sectors that the researchers and partners will work with? These questions served to emphasise the importance for all researchers, especially those from the Global North, to learn from Global South researchers' long-standing research and careful relationship building with knowledge holders and customary rights holders, with a view to revising assumptions and contextualising knowledge before starting new collective research.

WP0 also served to better understand researchers' expertise and contributions to better identify potential synergies within and across disciplines. It allowed to recognize which researchers had the knowledge, methodologies and trusted relationships to engage respectfully with communities in the co-identification in matters of concern; and who also had concerns about how One Ocean Hub research could create risks and negative impacts on these communities. It is important to note that we did not shy away from tension or conflict within the research team, and saw conflict and tension as a site for transformation and t-learning. Creating convivial events online and in person to hold and facilitate conversations on these tensions and conflicts was crucial. It also served to better understand the potential for co-learning from different communities and sharing the lessons among communities from different parts of the world through the Hub.

As part of WP0, we phased these convivial events across countries to offer insights for South-South learning across South Africa, Ghana, Namibia, and cross-regional learning. It was also critical to ensure the WP0 process was not facilitate in a didactic way, but for each research group, university and smaller collaborative team to interpret how and in what ways they might go about WP0 activities. As such, the scales to which each group conducted this varied, according to their capacities, networks and personal relationships on the ground. This required careful week-long 'inception' workshops in each country, with a diverse set of researchers from partner countries and other representatives to try to flesh out what WP0 would look like in each of their contexts.

Preliminary ethnographies were then carried out, through interviews with Hub Researchers to identify potential co-researchers within ocean-dependent communities, relevant research

expertise, gaps, and methods of engagement; interviews with ocean-dependent community representatives identified through previous research and interviews, leading in particular to collection of oral histories with communities, and desk-based historical and economics research. The preliminary ethnographies developed social cartographies (maps of knowledge(s), worldviews, concepts and struggles located into a cultural historical context) to understand matters of concern for all disciplines. These matters of concern were shared and distilled through a social learning process of 'call and response' from different people from various cultural, ideological, political contexts. These insights are particularly useful for: identifying questions, assumptions and misperceptions concerning ocean laws, economics and sciences; identifying opportunities in arts, cultural heritage and alternative non-extractive creative economies; and allowing the team to synthesize existing knowledge and key findings from previous research by researchers within the tam and other relevant researchers and knowledge holders in the country from all disciplines, where available. In parallel, legal research was carried on concerns within national legal frameworks identified in the inceptions workshops, which resulted in a more nuanced understanding of the extent to which "human rights and the environment" legal scholarship was lagging in connection with the marine environment and blue economy. It also served to reveal connections and mismatches between international human rights and environmental law research vis-a-vis environmental justice scholarship and practice. This served to provide a frame of reference to identify legal issues through the ethnographic work, connect local research with international law research; and identify other opportunities for Hub research to be shared with or contribute to relevant international governance processes that could support progress at the national and local levels (Morgera, 2024).

In-country working groups then reflected on the question, concerns, narratives raised by local participants in situ and reflect on existing research that can support these concerns, or problematize these. Preliminary findings from the in-country working groups, as well as from other researchers involved in the desk-based and virtual activities, were shared on a regular basis through an iterative and convivial process of creating online "living aulas" (Chaves and Wals, 2018). Living aulas were living classrooms where over the course of a year, we discussed how each research program could respond (by being adjusted, expanded or extended, or by clarifying its existing approach) to the emerging matters of concerns or 'ecotones' from the country teams for the One Ocean Hub. This created a relational cartography across different working groups and research programmes, allowing for new meaning-making opportunities across disciplines and ways of knowing to emerge.

We thus moved from thinking of Indigenous peoples, small-scale fishers, women, local communities and emerging researchers as "beneficiaries" towards engaging them as co-researchers through ongoing negotiation and expansive learning, in order to transform how we carry out research. Different researchers had relationships on the ground, and engaged research legacies, which influenced the degree to which participative parity and inclusion could emerge in each context. This itself was carefully navigated and an ongoing practice of trust-building, transparent communication on equity concerns and ideological tensions among researchers, and in some cases transparent budgeting and resource sharing across work packages was necessary. For example, certain Global South researchers had three decades of pre-existing relationships with small-scale fishers and displaced communities, so there were several questions to be explored before other

researchers could be "invited" into those relationships. In terms of capacity development, shifts to include emerging researchers in the co-design or leadership of research and capacity-strengthening initiatives provided new opportunities for more responsive, effective and mutually beneficial approaches. This in turn allowed for the emerging of new solidarity networks among researchers, decision-makers, and civil society, which themselves became new social processes to be studied and supported by social sciences expertise (Pereira and Erwin, 2023). This is increasingly recognized as necessary in environmental law research for avoiding oversimplification through dialogic knowledge co-production with activists and those with lived experiences of environmental injustices. Transformative potential, in particular, is identified in the engagement not only of "experts whose imagination is limited by the status quo" but of Indigenous peoples and local communities that "academic researchers should accompany" to amplify their voices, as embedded participants in the building of networks of solidarity (Burdon and Marel, 2017; Holder and Gillivray, 2017).

These transdisciplinary methodologies were in effect already a prototype that could also be replicated in governance processes (public consultation, prior informed consent, environmental impact assessments). In and of itself, this research approach contributed to make progress towards an overarching objective of the One Ocean Hub – "connecting disconnected dialogue" among key stakeholders, most notably vulnerable communities, women and youth, are not sufficiently involved in developing and implementing ocean management despite their historical, sometimes disrupted, ongoing but silenced, misunderstood, undocumented, or threatened contributions to ocean stewardship. This approach supported us in identifying any inequalities and stereotypes that derive from colonial legacies still shaping dominant conceptions of development, progress and conservation that are not compatible with Indigenous peoples' and local communities' understandings and cultural rights (A/77/290, 2022).

Arts and social sciences research, therefore, created new learning spaces where different ocean decision-makers and experts (including educators, children, EIA consultants) could discover unconscious bias and challenge one other to see different dimensions of justice at stake. In addition, these were spaces where it was possible to connect emotionally (through empathy and imagination) with different experiences of marginalization across generations (McGarry, 2014) from ocean decision-making and negative impacts on livelihoods and cultures due to decisions on the use or conservation of the ocean. This was particularly important for those researchers that did not have the privilege of working with and learning directly from Indigenous knowledge holders and local knowledge holders, but whose areas of research were still relevant to contribute to an integrated and co-developed knowledge base.

These networks concretely have led to significant science-policy changes, such as legal victories in national courts where Indigenous peoples and small-scale fishers successfully opposed unsustainable uses of the ocean that had negative impacts on climate change mitigation, marine biodiversity, as well as their right to food and cultural beliefs (McGarry, 2023). This was indeed a crucial moment in the life of the One Ocean Hub, where the iterative co-identification of research questions with Indigenous peoples and small-scale fishers with regard to increasing threats to their ocean-related ways of life and their spiritualities revealed an area of shared purpose with biodiversity and fisheries scientists and modellers, who had been working on the ecosystem

approach to fisheries and found out how their science could support local demands for environmental and socio-cultural justice. This also provided an opportunity for national courts to confirm the relevance of the One Ocean Hub's transdisciplinary research findings in terms of human rights to culture and food, as well as to participation in decision-making of the communities that had been co-developing knowledge with the Hub. The choice of some social and natural science researchers to support and engage in national litigation, however, did raise tensions with other researchers who were part of government bodies or had ongoing relationships of advice to government. The judicial decision in favour of communities led researchers to reflect on their role in differentiating among stakeholders (rather than treating powerful and harmful stakeholders equally with vulnerable ones who contribute to ocean stewardship in the name of scientific neutrality), and to offer support knowledge holders with less agency and power with customary and sustainable values as a matter of transformative science and protection of human rights. The judicial decision drew from collaboratively created artistic evidence (animated film, theatre script and radio play) as pieces of evidence created in solidarity with small-scale fishers and Indigenous knowledge holders through One Ocean Hub facilitation. The artworks where entire collaborative research endeavours unto themselves and created new translation protocols across onto-epistemological values of the ocean, (particularly intangible heritage) and linked them to livelihoods, tangible heritage and ecological protections demonstrating that arts were generative science/society/policy ecotone activities in and of themselves.

On the whole, WP0 contributed to another overarching objective of the One Ocean Hub – to empower sustainable and fair partnerships, so that decisions regarding the conservation of the ocean (in relation to both biodiversity and climate change) and its uses support fair and equitable sharing of benefits arising from research and governance that enhance the wellbeing, resilience and livelihoods of groups in vulnerable situations. It also proved that deep tensions among researchers can be extremely fruitful areas for transformative change, that support decolonizing science and evolving relationships with ocean-dependent communities.

Responding to inequities among ourselves: the One Ocean Hub's Code of Practice

WP0 also served to develop the capacity of the Hub and its methodologies to engage in self-critical, reflexive, iterative, diffractive and challenge-driven research co-development with small-scale fishers, so that we could all be better equipped to respectfully engage with human rights holders. To that end, we co-developed research ethics and compliance protocols as our Code of Practice. We drew on recent initiatives in or with the Global South from other research networks - the San Code of Ethics (South Africa) and the Global Code of Conduct in Research in Resource-poor Setting – as well as from academic research on fair partnerships by certain Hub researchers (Wynberg and Houck, 2014; Morgera 2024b). It was also critical to learn from and collect negative lessons learnt by partners in previous research collaborations, and prior research on fairness by Hub researchers. The Code was co-developed through national researchers' workshops (including sessions with confidential submissions of areas of concern or negative experiences from the inception of the Hub or other research collaborations), and five Online Living Aulas gathering around 40 researchers from four regions from May-July 2019. The Code of Practice was then incorporated by reference in the collaboration agreements under the Hub, thus becoming legally binding. The Code was considered a living document that we continued to discuss, review, and

improve upon over the years. Researchers had an opportunity to submit an individual, confidential questionnaire to reflect on the relevance and implementation practices of the Code of practice, every year.

The Hub's Code of Practice contained: a series of Principles as shared values to guide future decision-making, risk management and learning under the Hub; a section on Research Ethics combining 'do no harm' with 'ethics of care' (discussed below); Safeguarding – our preventative approach against exploitation, bullying or harassment, including a process for raising and addressing concerns; a Publication Strategy (including intellectual property) to ensure recognition across the collaboration; a series of commitments to build an enabling environment for Early-Career Researchers; and a Data Management Strategy as a basis for developing ethical data-sharing practices.

The Principles, in particular, contain some of the most transformative learning arising from the process of co-development of the Code of Practice. One principle refers to Integration, reflecting on the need to integrate research across different disciplines and knowledge systems in order to reflect the intra-connectedness (Barad, 2007) of the ocean. On the one hand, this principle expressed commitment that the outputs of the One Ocean Hub research should amount to more than the sum of the parts (individual research programmes). On the other hand, it also underscored the paramount importance of diversity across the Hub (including different development contexts), which should be explicitly engaged with, as a valuable resource for Hub research on connections and disconnects, rather than being "subsumed" in integration efforts.

Integration was complemented by the "Rosetta Stone" principle, whereby Hub researchers committed to contribute to inter- and trans-disciplinarity, while acknowledging that they may do so in different ways and to different extents. To that end, the Hub sought to support effective communication and mutual understanding across disciplines (and gradually across knowledge systems), without the need to translate one discipline into the terms of another discipline. This was seen as essential in order to avoid the risk of something being "lost in translation." Instead, the Hub committed to use different formats (eg, maps, modelling, different forms of art) at the same time, so that different researchers could interact with the format with which they felt more comfortable while their respective inputs could be reflected in other formats that other researchers preferred. In the light of the "Rosetta Stone" principle, Hub researchers were expected at a minimum, to: make a proactive effort to enable others to understand their research and to understand others' research; interrogate their own discipline (to identify blind spots and limitations), by engaging with data and criticism arising from other disciplines within the Hub; and where possible, (re)direct the research to respond to ocean-dependent communities' needs, which turned out to be particularly challenging for less flexible research activities, such as that in labs or research vessels that is usually subject to large upfront investments and prior, detailed planning over shared infrastructure and staff time. We thus agreed that researchers should explain to other co-researchers, including within ocean-dependent communities, why they were not able to respond to their requests with their existing resources, and explore whether in-kind contributions from existing partners, new partnerships or use of flexible funds, could still help respond to these needs.

Another principle enshrined in the One Ocean Hub's Code of Practice was Complementarity. Hub researchers were expected to engage in research activities funded by the Hub that were complementary and added value to those undertaken by Hub partners, as well as those undertaken outside the Hub (filling gaps, bringing together sectoral research, setting sectoral research into broader context). This may seem a relatively obvious principle, but the reality of the fierce competition over ocean research funding, particularly for deep-sea research cruises, and uneven or unfair practices in sharing research findings within and across projects due to other competitive dynamics in the sector, made this an ongoing area of research, iterative learning and relationship-building. In addition, it required creative thinking in the Hub's relationships with non-research partners, notably UN bodies and government agencies, with a view to understanding how research could fill gaps in areas that were beyond their mandates or considered as too politically fraught. The Hub thus contributed to connect existing research projects through the creation of enabling mechanisms to share research across countries, regions and scales (multi-scale syntheses), as well as through new funding applications and legacy activities.

Another key principle in the Code of Practice was Nourishing: Hub researchers committed to support one another through peer-learning and constructive peer-review, giving equal weight to the Hub's and the individual researchers' needs, in a context of conviviality, solidarity, openness and acceptance. This was also a principle that played a key role in supporting difficult conversations, by committing researchers to understanding respective vulnerabilities, learning from failure (rather than only celebrating successes), and engaging in attentive and active listening (ie learning through others' knowledge and experience). A related principle focused on recognizing Multiple Dimensions of Fairness. It concerned both fairness under the Hub, as well as multiple sides to an equity issue in the science/society/policy ecotone. This principle helped identifying collective approaches to fairness, by considering: groups in situations of vulnerability ("partnership" implies, for instance, being able to respond to the questions "when are you coming back?" and "how can we keep in touch between this visit and the next?"); fairness questions in each region as well as fairness issues across regions, where researchers operated; and among research partners (including previous trust-based relationships between specific partners and specific researchers under the Hub). There were also questions related to fairness in our relationship with ocean research funders and tax-payers. Within the Hub, the principle of multiple dimensions of fairness committed researchers to identify, understand and address differential power relationships – Global North/South, institutions with more or less budget, individual researchers with more or less seniority. One practical area to apply the principle was the recognition of the origin of ideas and contributions, through frank and explicit conversations, under the Publication Strategy, for instance.

While these principles were developed specifically to guide transformative ocean research, they provided relevant also in the context of ocean governance. For instance, they provided the basis for science-policy recommendations that an inter-disciplinary group of Hub researchers developed for the intergovernmental negotiations on Global North/South equity issues such as those of the Agreement under the United Nations Convention on the Law of the Sea on the Conservation and Sustainable Use of Marine Biological Diversity of Areas Beyond National Jurisdiction (BBNJ Agreement: Morgera et al, 2023), but also of the governance of digital sequence information under the Convention on Biological Diversity or the negotiations of pandemics treaty (Morgera, 2024b).

Notably, the Hub principles provided standards and practices confirming the need for an iterative, explicit process to address and learn from the tensions arising from seeking to address or ignore power inequalities in the context of knowledge production and science-society-policy integration processes. In other words, over five years, the One Ocean Hub practices based on these principles provided "proof of concept" that iterative, transdisciplinary and transgressive learning could be applied to other ocean research projects (notably as part of the UN Decade for Ocean Science, as the Hub was invited to become an Implementing Partners on transdisciplinarity), as well as a transformative approach within international decision-making processes.

Transformations in marine science: a deep example of inequalities and inequities in ocean knowledge making

During WP0 and in the development of the Code of Practice, we also had an opportunity to identify and start to address the specific inequalities and inequities in the distribution of ocean (particularly deep-sea) scientific resources and funding, which is a major and well-documented issue for global, regional and national ocean governance too. The capacity for ocean science and policy-making is often concentrated in the Global North, while the Global South has limited resources and infrastructure to support research and policy development. This global inequality is keenly reflected in the evidence of inequalities that arise from intellectual property applications, whereby only ten countries in the world appear to be benefitting from deep-sea research. Only 10 countries account for 90% of patents related to marine genetic resources (the US, Japan, certain EU countries, Switzerland and Norway: Arnaud-Haond 2011).

But other aspects of deep-sea research are equally in the hands of a handful of States. There is only a restricted number of countries that can afford the costs and risks of deep-sea research vessels and therefore can control who has access to that source of knowledge. As a result, the vast majority of developing countries are not part of deep-sea research efforts and are also greatly underrepresented in marine taxonomic research (Broggiato et al, 2018). This has vast implications for developing countries' capacities to prioritize key research questions that are essential for ocean management within and beyond national jurisdiction, as well as for their influence in international ocean policy processes (Morgera, 2024c) and climate-related decision making (Morgera et al, 2024b).

As Hub researchers involved in the Ocean Genome report indicated, growing Global North-South scientific collaborations are 'characterised by pharmaceutical or biotech companies working with established centres of excellence located in high-income countries' (Blasiak et al, 2020). As a result of these, as well as the increasing reliance on sequencing technologies and bioinformatics, 'the capacity to undertake genomic research …is inequitably distributed among countries' (Blasiak et al 2020). Thus, a call has been made to urgently 'promote inclusive and responsible research and innovation that addresses equity differentials and fosters capacity and access to technology, while facilitating the realisation of commitments to conserve and sustainably use the ocean's genetic diversity' (Blasiak et al 2020). There are also increasing calls to improve diversity and inclusion in deep-sea research (Amon et al. 2022) and to question the meaning, motivation and metrics of capacity development efforts (Harden Davies et al. 2022), that should be more clearly connected with efforts to decolonise ocean sciences. Without dedicated initiatives to redress these inequities,

the situation is likely to worsen, as increasingly sophisticated and technologically demanding methods proliferate with growing risks of further social inequality, genetic commodification and privatisation (Shen et al. 2024).

Understanding the ocean genome (eg the whole of the genetic material present in all marine biodiversity, including both the genes and the information they encode) is essential for 'determining the abundance and resilience of biological resources' and 'increas[ing] awareness of the pressures facing marine biodiversity', but we are far from having that understanding as the 'functions of some 90 percent of genetic sequences collected from viruses remain unknown' (Blasiak et al 2020). Deep-sea knowledge is foundational to sustainable use, effective conservation and management of marine areas beyond national jurisdiction (ABNJ), as well as in deep-sea areas within national jurisdiction, and in fact also coastal areas due to the ecological connectivity of the ocean that relies on currents and the movement of migratory species (Popova, 2019). So, States with limited ocean knowledge are going to be necessarily less able to participate in decisions on conservation and management in ABNJ, as well as less able to manage sustainably marine spaces within national jurisdiction, even if they are among those with strongest connectivity to ABNJ (Morgera, 2022). The specific challenges for Global South researchers arising from these inequities in marine sciences were a specific object of study under the One Ocean Hub (Lavelle and Wynberg 2022). Against this background, one of the key recommendations emerging from an inter-disciplinary assessment of barriers to deep-sea research in South Africa was the need for research partnerships that promote co-developed, co-led and co-published research, where mutual learning is a key factor from the planning stage (Sink et al. 2021).

In addition, as it became clear during WP0, the most inequitable budget allocation across the whole research collaboration was that concerning deep-sea research, as it involved the use of a UK deep-sea research vessel and allocated almost 90% of the budget to UK researchers who had the know-how and equipment to lead the research at sea. Access to the deep-sea research vessel and relevant know-how was essential to allow for the first time the sampling of seamounts and deep slopes and abyss in deep South African waters. On the one hand, there was little, if any, flexibility to renegotiate the budget allocation, due to how respective institutions had locked-in resources as a pre-condition for their participation in the project. On the other hand, it was at least possible to explore how the lessons learned from the recognition of such inequity (which is generally taken for granted in other deep-sea research projects) and from a frank– often, very difficult – conversation on the inequities led to the opening up of a process towards co-identification of research questions and approaches for the deep-sea exploration that would be beneficial both for researchers in the Global North and in the Global South. This was aimed at co-developing an approach to the cruise research planning process and its co-leadership. It would have entailed co-setting the research priorities and ultimate objectives, co-selecting the research team that would be on the research cruise, considering opportunities for exchanges of lab personnel to participate in different types of analysis, as well as co-selecting research paper leads and equitable contributions to these and other outputs. On the whole, this process allowed to identify concrete opportunities for sharing of know-how (for instance, Global South researchers being cc-ed in emails by Global North researchers ordering equipment for the deep-sea research cruise) that would have increased the capacities of researchers in the Global South to negotiate fairer conditions for future research

cruises and be better placed to co-lead them. While the co-developed Global North/South research cruise did not eventually go ahead (due to unforeseen budget cuts), the agreement on the need for a process to co-develop a deep-sea research cruise programme nevertheless provided important learnings that are applicable to other ocean science collaborations.

These exchanges also led to the idea of a dedicated capacity development cruise that was responsive to emerging researcher needs from seven different institutions in South Africa and Namibia. This is an example of South-South ocean science cooperation. The expedition was led by emerging researchers, utilizing new technology to reach unprecedented depths in the sampling of the marine environment to co-produce knowledge needed to address the challenges associated with developing sustainable oceans economies in South Africa and Namibia. This represented a different approach to capacity development from the common models where African researchers are invited to participate or virtually observe field operations, using technology that they are unlikely to access in their country (Adams et al, 2023).[3]

Other initiatives Global North/South cooperation included collaborations to advance reference images for more standardised image analysis, greater data set interoperability and comparability (Howell et al. 2019) and even machine learning application (Adams et al. 2024) and new collaborations to include southern partners in marine habitat and ecosystem classification work (Bridges et al. 2022, McQuaid et al. 2023). A new network of African Deep-Water Researchers was also catalysed to respond to the disparity in deep-sea research capabilities between the Global North and South. Established in 2023 under the UN Decade of Ocean Science-endorsed Challenger 150 Programme and led by an early-career researcher, the African Network of Deep-water Researchers[4] is connecting researchers, skills, infrastructure and resources within Africa, and with a global community of deep-water researchers, to build on the continent's capacity for deep-water research in support of sustainable development. To date, the Network comprises over 250 members from 27 African nations.

Beyond the equity issues across the academic disciplines involved in deep-sea research, the One Ocean Hub researchers working across marine deep-sea science and social and cultural relationality to the ocean, discovered powerful resonances and dissonances between Indigenous framings of the deep-sea and scientific renderings of these ecosystems. In South Africa, Namibia and Ghana, the role and value of the deep sea in spiritual and customary lore extends well beyond 'known positivist science' and into mystical and mythical meanings, cosmologies and worldviews that are entangled with the ocean and deep-sea ecosystems in particular. One Ocean Hub researchers also tentatively acknowledged the possible collaborative insights available across Indigenous and scientific understandings of the deep sea in identifying deep sea species and their associative value. Hub researchers found Indigenous healers and diviners are accessing deep-sea species surfaced into the national informal economy as by-catch from deep-sea fishing vessels. These species have entered the socio-cultural economy and the spiritual and mystical consciousness of citizens in the

[3] https://oneoceanhub.org/dedicated-cruise-deepens-offshore-research-capacity-in-south-africa/.
[4] https://oneoceanhub.org/new-network-of-deep-water-researchers-aims-to-build-africas-capacity-for-deep-water-research/

form of use in traditional ritual and or medicines or 'muthi[5]', as is the case in South Africa. Here deep-sea bio-cultural diversity is emerging and interacting with deep-sea science. These parallel research communities (both Indigenous and Western scientific) could have meaningful contributions to health protection, including in the context of future global crisis such as COVID19. In this context, deep-sea knowledge is emerging in parallel in very different contexts and using very different meaning-making and knowledge making frameworks, the outcomes of which can both support and influence modern medicine. Some researchers are recognizing and affirming these resonances, with One Ocean Hub collaboration, making space not just for marine scientists but cultural practitioners and healers. Beyond the material resonances in the context of medical bio-prospecting, there are spiritual/scientific resonances as well. This goes further, when considering the violent and unjust history of South Africa, where the Witchcraft Suppression Act 3 of 1957 not only dismissed Indigenous marine knowledge in relation to medicine and healing, but systematically and violently excluded people from the ocean.

This was most noticeably visible in a playful and response-ablely fashioned short film by South African Hub researchers (Marine Scientists/Artists/sociologists and others) entitled "Indlela Yokuphila: The Soul's Journey". The animation traces a Zulu story of how the soul travels into the afterlife (which exists in the hydrosphere), revealing two very different ways of framing the hydrological cycle (scientific and Indigenous) and the role of the deep sea. The animation powerfully illuminates how Western scientific and Indigenous framings surprisingly align. This is a direct example where collaborative T-learning research, led by arts-based meaning-making created knowledge translation protocols across Indigenous and Western Scientific modes of meaning-making, that in turn could be metabolised by a legal process, and challenge evidence hierarchies and what counts as credible knowledge in a court of law (See McGarry, 2023 for more on this). Ultimately, the animation showcases multiple ecotones in the transitions between death and life, Indigenous and western scientific knowledge, and across the many ecosystems from the terrestrial and freshwater realm to the deep sea.

The tyranny of marine science that remains unconscious

In this section, we complete our reflection on decolonizing ocean science through transdisciplinary and transgressive research with a focus on the role of Western science in the context of fortress conservation and discrimination, and gender inequality. We also reflect on the crucial role of ethics of care to navigate these tensions towards generative spaces for transformation. This is part of ongoing and iterative efforts to learn from the mistakes of the past. Here we focus on transgressive research capacities to not only practise academic rigour, but political rigour too (Temper et al, 2019).

In the context of the One Ocean Hub research, we have wit(h)nessed the negative consequences of so-called 'neutral' or 'objective' marine governance that has influenced the fortress conservation

[5] "Muthi" is a term used in southern African traditional medicine and refers to various plant, animal, and mineral substances that are believed to have healing properties or spiritual power (Dold and Cocks, 2002). Muthi is used for a variety of purposes, including healing illness, warding off evil spirits, and promoting good luck or prosperity. The use of muthi is often deeply embedded in cultural and spiritual beliefs, and it continues to be an important part of many people's lives in southern Africa

models dominating protected areas in coastal territories. Territories that were subject to traditional fishing rights were rightfully won back in successful land claims. Despite this land technically belonging back to local people, the very strict management systems inherited from colonial rule continued to dominate conservation management practice, which are shaped by an untransformed biodiversity science. This led to the killings of Nibela fishermen in South Africa, by Marine Protected Area officials who have been trained to see biodiversity conservation without adaptations to accommodate customary forms of marine governance, traditional and intangible heritage and other approaches to co-management, despite legislation demanding all MPAs should practice co-management. These blind spots are further reinforced by legacies of apartheid and structural racism embedded in the conservation sector. In both South Africa and Namibia, the legacy of apartheid policies, gentrification/tourism and coastal mining and coastal protection efforts have displaced countless communities away from the ocean, and stripped them of their ocean heritages and livelihoods. Historically western sciences such as eugenics justified slavery and racist policies, and with the first slave port in the cape coast of Ghana, now today is suffering from wage and child slavery within the fisheries sector, due to incoherent marine policies built on only part of the story. Eugenics also led to racist policies such as apartheid, which violently restructured peoples' relation to land, their knowledge and their own culture. The legacies of apartheid in South Africa and Namibia, and the ways these were tied up with conservation agendas and economic plans, have savagely dispossessed black and brown bodies of their relationships and livelihoods with the sea.

In 2024, the American Society of Human Genetics (ASHG) issued an apology for the involvement of its early leaders in the eugenics movement and for the organisation's failure to recognize and oppose past injustices in the field of genetics. The apology comes after a yearlong project by ASHG that resulted in a 27-page report documenting instances of injustices, including ASHG leaders who supported forced sterilisation and the organisation's silence on the use of genetics to justify discrimination against Black people. The ASHG report states that the findings are "painful" but must be widely shared, and argues that expressing remorse and condemning past inappropriate actions is necessary to build trust. This systematic, attentive, conscious and reflexive turn by genetics scientists, is leaning towards practical acts of practising 'science-with-a-conscience', and not just trying to 'do-no-harm'. Apologies and conscious acts of reparations in marine sciences and their role in past and ongoing exclusionary marine conservation are an essential step in decolonizing science, and can be meaningfully facilitated also with art-based research approaches.

While there is increasing awareness of the specific gender inequalities in ocean science, including documentation of gender-based violence and discrimination in the fisheries sector (Mangubhai et al, 2023), there is a need for a deeper engagement within feminist ecological and post-qualitative research that offers a variety of theories, methodologies and ethical approaches that can support more responsible and accountable marine research forms within transformative and transgressive knowledge co-production. Critical African, post-qualitative and other feminist researchers have highlighted how science has historically been shaped by patriarchal colonialism and empire-building, and has become blind to the embodied, relational and more-than-human entanglements that have led to the creation of contemporary marine science (Haraway, 2016; Ellingson, 2019; McKitrick 2020; Gumps, 2021).

This needs to be taken into account in global research partnerships, as universities in the Global South still suffer from historically western approaches to teaching and learning, and this in turn influences how research is conducted and communicated. Wane (2008: 193) argues that teaching and research within African universities occur amidst a historical background of colonialism, imperialism, and various forms of post-colonial and anti-colonial conditions. This environment with its eurocentric influences further influences the ways in which we teach sciences, and our knowledge-making in general (Lundgren and Prah 2010; Moletsane 2015; Knowles, 2022). One such South African post-humanist feminist researcher Viv Bozalek (2020) calls for "Rendering each other capable"... and "doing response-able[6] research responsibly". She forefronts the need to move beyond the limitations of using principle ethics in research practices, which are premised on foundational notions of 'do-no-harm' (Bozalek, 2020). Instead offers ongoing care ethics, which can inherently shift the ways in which research is practised, and for whom, while at the same time exploring how researchers can remain accountable and response-able.

Bozalek (2020) proposes a posthuman political ethics of care as an alternative. This approach emphasises the importance of response-ability, accountability, curiosity, attentiveness, and ultimately rendering each other capable in research. Care is essential in the process of "rendering each other capable," and in responsible collaborative research it involves empowering individuals and communities to take control of their own lives and circumstances, including the ways in which knowledge is produced and how, and for whom. Bozalek argues that care is not just a basic human need, but also a crucial factor in achieving social justice and equity. She explores how caring relationships can foster agency, resilience, and social transformation. Bozalek's work highlights the importance of acknowledging and valuing care in our personal and societal lives, as it is a vital component in creating a more just and equitable world. She uses post-human physicist Karen Barad's (2007) relational ontology that posits that entities do not pre-exist their relationships, but rather come into being through relationships. This means that agency is not a property of an individual, but rather a performance within a relationship, and that matter is a congealing of agency (Bozalek, 2020).

This perspective makes the notion of the 'individual' as more entangled and relationally constituted than we have been led to believe, and as such emphasises that nothing can be considered outside of relationships. The idea of ethical care practices is also co-constituted through relationships. However, this view is not in line with the established normative framework of principle ethics[7] used by university ethics committees to make judgments about research. An ethics of care, that

[6] Bozalek (2020) uses the concept of response-ability in research which is inspired from Donna Haraway's (2016) idea of playing into abilities to respond in which researchers allow themselves to take chances, be joyful, improvisational, and inventive in their research. Response-ability involves becoming-with, making together, and rendering each other capable (Bozalek, 2020). It requires a new way of being obligated, through response-ability between human and non-human beings, that promotes flourishing or living and dying well in the world. This approach should take seriously the possibilities of becoming-with, making together, and rendering each other capable of engaging more meaningfully with research respondents.

[7] Principle ethics are the most established approach to health and social science ethics, based on rights-based and traditional social justice theories. They propose four principles for moral reasoning in health care: respect for autonomy, non-maleficence, beneficence, and justice (Beauchamp, 2010; Beauchamp & Childress, 2013). However, according to Walker (2008) and others (Bozalek, 2020) there are issues with relying solely on these principles for ethical practice, as they prioritise the rights of the individual and tend to favour universal rules. An alternative view is the expressive-collaborative model of ethics, which focuses on responsibility and social practices.

forgrounds relationships, relationality and the social tissue between researchers, policy makers, and Indigenous knowledge holders, allows for a shift in research parity and more equitable and inclusive knowledge co-production. This was significant in the framing, formulation and practices of a transdisciplinary hub such as the One Ocean Hub, as it centred social relationships, avoided assumptions that agency is individually constituted and saw friendship, understanding, empathy, attentiveness and even imagination as important indicators of impact/agency in measuring transformation in the science/policy ecotone.

In the experience of the One Ocean Hub in the past five years, we saw progress in such reflexivity of marine scientists who are expanding beyond no-harm towards ethics of care, and beyond academic rigour, to incorporate political rigour (Temper et al, 2019) in how they frame, design and carry out their research. Returning to our intertidal figuration, we are wit(h)nessing tidal shifts between disciplines, where fluid knowledge pluralities are spilling over from one disciplinary tidal pool to the next. We believe that acknowledging and addressing the role that marine science and policy has played in perpetuating systems of oppression and violence, is a vital step in building trust, situated solidarities and meaningful change towards inclusive ocean decision making. This requires a commitment to decolonize science, expand Indigenous and plural approaches to knowledge co-production, challenge evidence hierarchies, and address issues of gendered power dynamics and global inequalities. Such an effort entails a commitment to transformative action and a willingness to challenge existing power structures in universities, government agencies and in decision-making theatres that design policies. Transgressive and transformative praxis in ocean knowledge co-production is vital in addressing the inequalities and inequities that exist in marine science.

Surfacing and navigating tensions means researching them – not seeing them as things to be ignored or to be 'fixed' immediately, but as generative thematic areas of work that can enrich how we can learn together and discover opportunities for transformation both in research and in governance. For instance, Hub researcher collaborations that surfaced and explored injustices arising from inadequate participation in the design, implementation and management of marine protected areas, led to transformative collaborations where marine scientists called for social sciences-led and art-based approaches to re-imagine participatory planning of new marine protected area designations that include Indigenous knowledge and contribute to recognize intangible ocean cultural heritage, recognize traditional fishing rights and protect human rights.[8]

FROM TRANSDISCIPLINARY AND TRANSGRESSIVE OCEAN SCIENCE TO TRANSFORMATIVE OCEAN GOVERNANCE?

This chapter has explored why it is important to see ocean knowledge as plural and allowing for tidal shifts between these, as transdisciplinary and transgressive ocean research is a precondition (and in fact part and parcel) of transformative ocean governance. While this work is ongoing,

[8] https://oneoceanhub.org/connecting-science-policy-recommendations-across-ocean-research-projects/.

rather than trying to conclude something that is intertidal and in a constant state of flux, we chose to share some key learnings and recommendations in leu of a conclusion.

Key learnings and recommendations include:

1. Apply inclusive transdisciplinary research methods to build the relationships, knowledge (SDG14.a) and capacities for transformed ocean governance, contributing to SDG17 on partnerships. The intertidal knowledge landscape is generative: the interface of different knowledge systems, including Indigenous knowledge, small-scale fishers' knowledge, other local knowledge is an exciting avenue for research that opens paths towards transformative change.

2. Create a Work Package Zero to co-define concerns, gaps and disconnects, and to co-design research questions and a range of methods with Indigenous Peoples, small-scale fishers, and women. People and context shape research: recognizing that social sciences (sociology, education, human geography) should lead the co-design phase and can help identify benefits in realising not only SDG 14 but also SDGs related to poverty, food, health, cultural heritage and climate change (SDGs 1-3, 11 and 13).

3. Learn from past mistakes and injustices (within your discipline, in other research collaborations, from prior research in the Global South) and co-develop a Code of Practice, including ongoing and iterative Ethics of care (going beyond no-harm).

4. Don't be afraid of tensions. Transdisciplinarity is hard, but the tensions can be valuable generative sites for research and novel approaches. Tension in the strings of an instrument that are necessary for it to make music. So tensions can create resonance: focus on identifying tensions as areas of learning and knowledge expansion/accountability - including areas of inequity.

5. Embrace the process of decolonizing science and don't underestimate conviviality, friendship building and relationship building in your research.

6. Recognize artists as researchers: Arts is a research discipline, not just a means of communication. Artistic endeavours can mediate and support boundary-crossing work between disciplines and world views/cosmologies, and can research blind spots in contemporary science.

7. Co-produce research with policy makers at different levels: they are not just an audience. Research co-production done with policy makers can prototype new modes of inclusive and fair ocean governance, contributing to protecting ocean-dependent communities' human rights and everyone's human right to a healthy environment (SDG 16).

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